Decoys That Do Work: Evaluating Executed Cover for LLM Agent Communication Privacy

***Abstract*—Decoy tool calls change both an LLM Agent's encrypted traffic and the work its services perform. We study how call generation affects task recognition and measure inference, goal completion, business-state integrity and cost on the same executions. Our 3,000 single-host executions use two fixed-weight planners over MCP and TLS 1.3/HTTP/2. Controller-inserted cover bypasses the model wait preceding ordinary calls, exposing a short-gap signature. An exploratory reanalysis shows that time remains useful after waits are sampled from training traces: removing time features reduces selected-attack recognition in three of four history-dispatch cells after Holm correction, including 99 to 66 and 106 to 77 of 120 concurrent tasks. Context removal and joint decoding also recover recognition without supplied test call windows, while retaining episode association. They improve over refitted baselines in five of eight cells after Holm correction; concurrent scores rise from 56 to 80 and from 78 to 112/120. Neither shadow cell improves with the added attacks. Shadow cover instead has its own complete-policy trade-off: more and different calls, about twice the model tokens, and residual record-only maxima of 73 and 69/120. These results connect cover generation to observable trace structure and motivate evaluating its privacy, business authority and resource cost together.**

1. Introduction

An LLM Agent receives a support ticket, checks an order and submits a refund. An observer of the encrypted path to the commerce service sees contacted endpoints, exchange sizes and waiting times. Recent Agent studies use such metadata to infer prompts, tasks, workflows and user traits [2], [3], [4], [9]. Adding decoy calls can obscure the requested action, but each extra exchange must be generated somewhere and terminated somewhere. These choices determine its traffic signature and business cost.

The mechanism. In the planner loop we measure, an ordinary tool call follows a model decision. A controller can insert a decoy directly after a tool response, bypassing the next model wait. That shortcut separates the decoy by its inter-call gap. Delaying the call changes this feature, but leaves its destination, position, size and relationship to the other calls available to an attacker. A complete shadow task changes those relationships as well as adding a second planner's work. We compare these implemented choices on actual TLS records and track the work performed with each trace.

Classical query obfuscation already links decoy quality to its generator [63], [64], [65]. Our question is how that principle interacts with an Agent's model decisions and tool authority. Immediate cover is therefore a calibration arm, not a proposed defense: cover that ignores the honest timing law is known to be separable [69], [64], and we include it to confirm that the attack suite detects this in a live Agent loop. A byte-only attack can miss the controller's timing shortcut; a timing-only explanation can miss remaining call structure; a goal-only check can miss collateral business effects. The measurements therefore pair observation information with inference, task outcome, business state and resource use.

Setting. We compare four cover policies and no cover in 3,000 live executions of Qwen3:8B and Llama3.1:8B over MCP and TLS 1.3/HTTP/2 on one host. The provider cooperates by padding responses and permitting isolated execution; history-based policies also use server-side minimum holds. Section 4 states these capabilities. When only request shape is private, protocol padding may suffice. Hiding response activity calls for additional responses, which a cooperating provider can synthesize or an application call can produce. Reaching the application need not require a business write.

Findings.

- **Time supports recognition even after sampled waits.** Immediate insertion exposes a model-wait shortcut. After sampling waits, an exploratory feature ablation still finds lower recognition without time in three of four history-dispatch cells after Holm correction. The concurrent comparisons fall from 99 to 66/120 (Qwen) and 106 to 77/120 (Llama). Qwen sequential remains nonsignificant after this family's correction ($p = .0931$). Explicit-position ablations show no significant difference in their separate family.
- **Context decoding recovers recognition from associated encrypted records.** With test call windows withheld, context removal and joint decoding improve over refitted baselines in five of eight exploratory comparisons after Holm correction. Concurrent recognition rises from 56 to 80/120 and 78 to 112/120. The original short-gap remover underperformed on this view; the stronger results show why an unsuccessful removal pipeline cannot stand in for a privacy assessment.
- **Shadow cover changes a complete policy's inference–cost trade-off.** The strongest tested call-

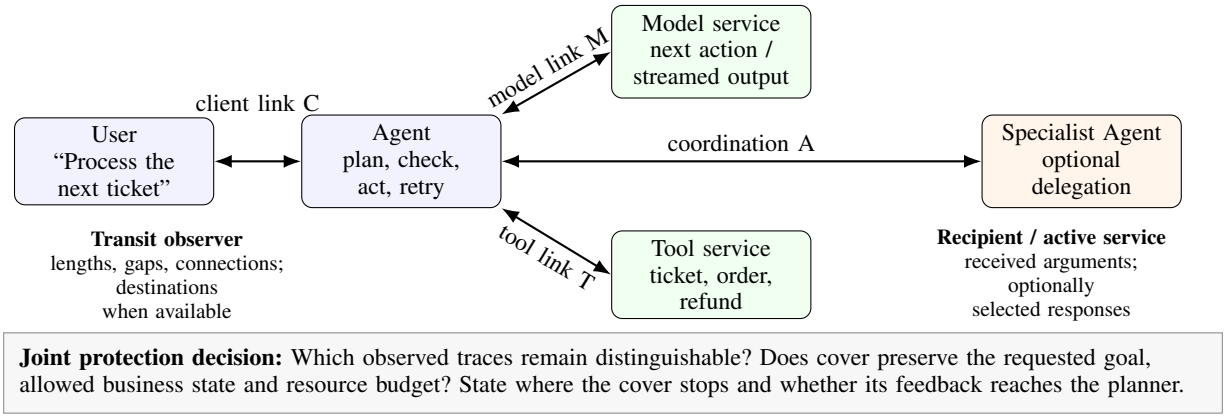


Figure 1. Running support-ticket example and observation rights. A tool result returns to the Agent and can enter the next model request. Each encrypted link exposes its own traffic shape; a recipient additionally sees the request addressed to it. Concurrent tickets may share a connection on either link. The arrows represent application communication, not a one-to-one mapping from tool events to packets.

window attack scores 70/120 and 64/120 at about twice the model tokens; record-only maxima are 73 and 69/120. Added context decoders do not improve the selected baseline in either shadow cell. The second business task changes call count, composition and terminal coverage together, so the comparison does not isolate online planning. Task, state and resource outcomes accompany those same executions; a separate stress control illustrates collateral effects missed by goal completion.

Contributions. We contribute (1) live measurements and feature ablations that distinguish the model-wait shortcut from residual temporal context; (2) context and joint-decoding attacks on records without test call windows, alongside complete-policy inference, task, state and resource outcomes; and (3) an execution-aware evaluation method that states cover termination, state and feedback authority, and service support. The new analyses reuse the measured executions and are exploratory. Section 5.3 provides an ideal candidate-label reference for interpreting shadows, and Appendix A supplies the forty-four-setting background.

2. Background and Motivation

2.1. Agent communication

An Agent uses an LLM to choose actions, and external feedback can change its later actions or termination. Figure 1 shows the links of the running example: client–Agent, Agent–model, Agent–tool and inter-agent coordination. A tool result returns to the Agent and can enter the next model request. Tools are commonly reached through the Model Context Protocol (MCP), either over local stdio or over Streamable HTTP [23]. Remote connections typically use TLS 1.3 and HTTP/2, which encrypt content but expose record lengths, directions and timing, and multiplex several request streams on one connection [25], [24]. A captured packet, TLS record, HTTP stream, tool call and complete task are therefore different units, and a boundary supplied by an evaluator is prior information, not an attacker output.

2.2. Communication side channels in Agents

A small but growing literature shows that Agent traffic leaks. Transit observers recover research prompts and user traits from contacted domains and timing [2], workflow classes from inter-agent communication graphs [3], and tasks or misuse from model-link and egress traffic [4], [9], [1]. Contacted recipients fingerprint Agent clients and models from protocol headers, terminal commands and tool-call arguments [6], [7], [48], [47], [35], and active defenders plant lures for Agent adversaries [10], [51]. The companion matrix maps all forty-four settings from fifteen such studies by observer access, target, unit and intervention. Most evaluate recognition; few evaluate a protection.

2.3. How traffic defenses are evaluated

Website-fingerprinting (WF) work has established what a defense evaluation must contain: realistic observer information and splits, attacks adapted to the protected distribution, and overheads measured on the same traffic [15], [16], [32]. Adaptive-attack and ML-security guidance make the same demands [19], [52]. Classical defenses pad, pace or add dummy traffic, and their cover never reaches an application [17], [18], [66], [34]. State-aware Agent benchmarks check environment outcomes [60], [61] but do not measure traffic.

2.4. Motivation for joint outcome records

Table 1 applies those requirements to the four Agent query/workflow privacy settings we located. WRA's five research decoys lower query recovery from .770 to .565, but its report-utility scores of 8.55 and 8.54 belong to a different intervention that changes destination visibility [2]. The communication-graph study refits its attacker on live

OHTTP captures and reports about thirteen-fold bytes, but no original-task outcome under that intervention [3]. No setting reports both an adapted attacker and matched utility, and the only executed-cover setting reports no matched utility at all. This is a statement about four settings from two sources, not a field-wide frequency; it shows that the measurements a deployment decision needs have not yet been made together.

3. Threat Model

Assets and adversary. The private target is the user's requested task, here one of six support actions. The defender controls the Agent and its controller. Tool providers are independent actors; the evaluated provider explicitly supplies the capabilities in Table 5, without granting general service control. We consider three observers, distinguished by their rights. A *transit observer* sees connection destinations, directions, ciphertext lengths and times on the links it watches. A *recipient* additionally reads the requests addressed to it, including arguments. An *active auditor* also chooses probes. Identifying a requested action does not establish whether it was authorized; that needs the principal and policy context. Table 2 maps these rights to common deployments and to the existing evidence for each view.

Where cover can terminate. Transport cover ends at an endpoint the defender controls, after the observer and before the application [34]. A relay outside a monitored egress can serve as that endpoint, but the hop from the relay to a third-party service remains. On that hop, the service's own protocol stack still offers two terminators the defender can use without the provider's cooperation. TLS 1.3 lets the sender pad any record and send padding-only records, which the receiver's TLS layer discards [25, Sec. 5.4]; HTTP/2 lets the sender pad DATA and HEADERS frames and send PING frames, which the receiver's HTTP layer consumes [24, Secs. 6.1, 6.7]. These mechanisms can shape the defender's own request bytes and timing. A padding-only record does not request an application response; PING requests a protocol acknowledgement with an eight-octet payload. Neither mechanism by itself specifies an application-shaped response or its service latency.

Cover termination therefore depends on the private signal and the provider interface. Request-side padding leaves application response activity to be addressed separately (Table 3). A cooperating provider can synthesize padded, delayed responses without executing the business operation. Where that interface is unavailable, additional application calls are one way to generate response activity. Our experiment compares calls executed on isolated clones; it is not an impossibility result for all other defenses. A recipient that operates the service sees every exchange it receives, so against it cover must also carry plausible arguments. Reaching the application does not by itself require a business write: a read-only or no-op call elicits a real response. Cover must perform a state-changing operation only if such operations produce responses or latencies that reads cannot imitate. We call any cover that reaches the application *executed cover*; the design space in Section 4 separates how much business work it performs.

Observer in this paper. Our measurements use a transit observer on the final hop to seven tool services, with no defender relay in between, and a privacy target that includes the exchange sequence and response timing. It receives destinations, ciphertext bytes and TLS record completion times. Section 7.1 additionally supplies episode and call windows, an evaluator privilege that Section 7.2 withdraws. The recipient view, where decoy arguments are also visible, is part of the threat model but not of these measurements (Section 8). The adversary may train on traffic generated under the same cover policy; the attack suite therefore includes classifiers and cover-removal models fitted on protected training executions, alongside attacks frozen on unprotected traffic.

4. The Executed-Cover Design Space

Four axes. We classify a cover policy along four axes (Table 4). *Termination* and *state* determine what business work cover performs: a disposable copy, read-only access, an isolated clone or production state. *Feedback visibility* determines whether cover responses enter later model requests and can change the real planner's decisions. The *dispatcher*, the process that decides when cover is sent, determines whether cover inherits the timing of honest calls. Honest calls follow a model decision; a controller that inserts cover without waiting for the model creates a timing signature that the first three axes do not capture. Our live study varies the dispatcher while holding termination (an isolated clone) and feedback (hidden) fixed; an earlier state control varies termination and feedback.

The execution contract. Because executed cover performs work, a policy must satisfy a contract as well as a privacy target. The contract names the objects and transitions cover may touch, the quota, token and latency budgets it may consume, and the feedback it may expose to the real planner. Isolation prevents direct writes to production state but still consumes compute and can change scheduling; a response hidden from the planner does not undo the operation that produced it. A requested goal can complete while another order, a shared quota or a ledger is wrong, so the contract separates three outcomes: requested-goal completion, violations of declared business invariants, and resource use. For a synthetic fixture whose intended transition is fully specified, exact state equality is the invariant; in production, invariants must admit legitimate concurrent updates.

The service contract. Every policy also needs capabilities from the service (Table 5). Our live study gives the defender a *cooperating provider*: services pad response bodies, hold a cover response for a sampled minimum duration, and execute cover on an isolated clone. This matches a provider that offers a privacy mode or a sandbox on the same endpoint, while its network path is observed by others. Such a provider could also discard cover in its application and return padded, delayed dummy responses, which is the proxy-discard row of Table 4; our dispatcher result applies

TABLE 1. EVIDENCE LOCATED IN THE RETAINED RECORDS. COUNTS USE ELIGIBLE SETTINGS; THE SECOND COLUMN ALSO GIVES DISTINCT SOURCES. "NOT LOCATED" LEAVES EVIDENCE UNRESOLVED WITHIN THE INSPECTED SCOPE. THESE COUNTS DESCRIBE AVAILABILITY, NOT STUDY QUALITY OR CHECKLIST VALIDITY.

Question	Eligible settings / sources	Located	Supporting settings and consequence
Protected-condition attack adaptation	4 / 2	2/4	Communication graphs: transformed captures and live OHTTP. Both come from one preprint.
Original-task utility under the same intervention	4 / 2	1/4	WRA visibility comparison. It does not price WRA's executed-decoy condition.
Adaptation and original-task utility in one setting	4 / 2	0/4	Adaptation and utility are located in different records; transport cost alone does not fill the task-outcome field.
Original-task utility for executed cover	1 / 1	0/1	WRA's five research decoys report recovery and extra research; matched original-task utility is not located.

TABLE 2. DEPLOYMENT AND ACCESS MAP. THE CONFIGURATIONS ARE ANALYST-DEFINED; CITED SETTINGS TEST THE LISTED VIEW, NOT EVERY COMPONENT OF THE DEPLOYMENT. A GATEWAY'S PLAINTEXT ACCESS REQUIRES TLS TERMINATION.

Deployment	Observer and permitted view	Existing evidence for that view	Protection consequence
Enterprise egress	A transit monitor sees routed model/tool flows, sizes, times and any exposed destination association.	WRA: supplied research-domain traces; ANT and cloud benchmark: model/combined traffic [2], [9], [4].	Specify which services cross the monitored egress. A union of flows is available only when the monitor can observe them.
Shared gateway or relay	Transit sees multiplexed gateway output. A terminating gateway can additionally read addressed requests.	Live OHTTP: passive-link inference under stated relay trust [3]; PrivacyTrace: semantic recipient/gateway access [35].	Separate relay association protection from content trust. Connection sharing changes the observation unit.
Local model, remote tools	An external observer sees remote tool traffic; the local model call has no corresponding external network leg.	WRA's destination-recovery mechanism is relevant to tool egress [2]; whole-deployment protection is untested in the mapped comparison.	Evaluate tool and downstream calls. Do not import a remote-model attack score into an absent network leg.
Cloud model, remote tools	An authorized vantage may see either or both encrypted legs. A contacted tool sees its own arguments.	ANT/cloud benchmark test model traffic [9], [4]; PrivacyTrace tests recipient-visible representations [35].	Check permitted model and tool views after the intervention; recipients and passive monitors need different protections.

TABLE 3. WHAT EACH TERMINATOR HIDES FROM AN OBSERVER OF THE FINAL HOP TO A THIRD-PARTY SERVICE. PROTOCOL-LEVEL PADDING IS DISCARDED BY THE SERVICE'S TLS OR HTTP/2 LAYER WITHOUT ITS COOPERATION [25], [24], BUT RESPONSES STILL FOLLOW REAL APPLICATION CALLS.

Cover mechanism	Terminated by	Still visible to the observer
Defender tunnel or relay	Defender endpoint	Everything on the relay's upstream hop
TLS record padding, padding-only records	Service TLS layer	Response presence, size and latency of real calls
HTTP/2 frame padding, PING	Service HTTP/2 layer	As above; PING acknowledges an 8-octet payload
Read-only or no-op call	Application	Read-versus-write response differences; quota use
Business operation (sandbox or live)	Application	Only what the attack can still separate

to that design too, because it concerns when cover requests are sent, not what the service does with them. A non-cooperating provider offers none of the service-side capabilities: the observer additionally sees unpadded response sizes and unheld response timing, and cover executes against real state. We therefore read our leakage results as measured under defender-favorable service conditions, and we do not measure the non-cooperating case.

Where executed cover sits among defenses. Name hiding and relays change destination association; padding changes lengths; pacing changes timing [17], [18], [66]. These layers remain useful, and Table 3 shows when transport cover suffices. When response activity is private, cover responses are needed as well as cover request bytes. A cooperating provider can synthesize them at an application terminator; otherwise the service interface determines which calls can produce them (Section 3). Model services add their own signals, such as token streams, cache behavior and token counts [20], [12], [11], [13], [21]; a tool-side cover policy does not shape later model requests. The evidence archive indexes the full layer comparison (Supplement S4).

TABLE 4. COVER TAXONOMY. EACH POLICY IS CLASSIFIED BY WHERE IT TERMINATES AND WHAT STATE IT TOUCHES, WHETHER ITS RESPONSES REACH THE REAL PLANNER, AND WHICH PROCESS DISPATCHES IT. THE LAST COLUMN NAMES THE EVALUATION IN WHICH THE POLICY IS MEASURED; S1 IS THE EARLIER STATE CONTROL AND SECTION 7.1 THE LIVE TIMING STUDY.

Policy	Terminates at / state touched	Feedback to planner	Dispatcher	Measured in
Transport cover	Tunnel endpoint before the service; no business state	No	Sender schedule	STP [34]; WF padding [66], [18]
Proxy discard	Defender proxy computes on a disposable copy	No	Controller	S1
Read cover	Real service; reads only	No	Controller	S1
Isolated execution	Real service on a persistent clone	No / yes	Controller	S1 (hidden and visible); all live cover arms
Live write	Real service on production state	No	Controller	S1 (stress control)
Research decoys	Real third-party Web destinations	Not located	Decoy generator	WRA [2]
Immediate	Isolated clone	No	Controller, directly after an honest response	Section 7.1 (calibration)
History-matched	Isolated clone	No	Controller, training-gap and duration samples; sequential or concurrent	Section 7.1
Planner shadow	Isolated clone; separate planner context	No	Second planner on an independent ticket	Section 7.1

TABLE 5. SERVICE CAPABILITIES USED BY THE LIVE STUDY AND WHO MUST PROVIDE THEM. THE STUDY THEREFORE MODELS A COOPERATING PROVIDER; AGAINST A NON-COOPERATING PROVIDER ONLY CLIENT-SIDE CAPABILITIES REMAIN.

Capability	Used by	Provided by	Non-cooperating provider
Response body padding (16 KiB quanta)	All arms	Service	Unavailable
Server-side hold of a cover response	History arms	Service	Unavailable; client can delay only its next request
Isolated clone for cover execution	All cover arms	Service	Only with a sandbox on the same endpoint
Request padding, dispatch timing	All arms	Client	Available
Same-service serialization	All arms	Client	Available

TABLE 6. JOINT EVALUATION CHECKLIST FOR ONE INTERVENTION AND POPULATION. ATTACH EVIDENCE OR STATE THE MISSING ITEM; GOAL COMPLETION AND WHOLE-STATE CORRECTNESS ARE SEPARATE FIELDS.

☐	Hidden target, population, prior or base rate
☐	Allowed observations and supplied boundaries
☐	Cover termination, privileges and isolation
☐	Planner-visible cover feedback and dispatcher
☐	Declared tolerances: inference, goal, state, tokens, bytes, latency
☐	Adapted attacks and largest tested residual
☐	Original-task attempts, completion and failures
☐	Business invariants and violations
☐	Matched bytes, latency, tokens and quota

5. Evaluation Methodology

5.1. Joint evaluation protocol

Privacy, task success, business state and cost must refer to the same intervention and evaluation population. The protocol has four steps; Table 6 is its reporting form.

1) **Specify observations and execution authority.** Name the hidden target, the visible links, recipient rights and any supplied boundaries. State where cover ends, which state it can access and whether its responses reach the planner.
2) **Specify the contract and tolerances.** Define the requested goal, allowed collateral transitions, and the inference, quota, token and latency tolerances, before measuring.
3) **Measure matched executions and adapted attacks.** Pair baseline and policy on the same workload. Include cover-removal and schedule-aware attacks, report the strongest tested residual inference, and take task, state and cost outcomes from those protected runs.
4) **Assess requirements.** Fix the attack selection and uncertainty procedure before evaluation. Joint acceptance requires simultaneous support for every mandatory endpoint; an unresolved endpoint leaves the policy undecided. Record forbidden executions separately from goal completion and compare observers using the same attack suite. Our worked example is descriptive because full joint uncertainty is not implemented.

5.2. Attacks and statistics

Each condition has its own attack suite, fitted on its training split. Size-only attacks are logistic regression (LR) and a 300-tree random forest (RF) on destinations and ciphertext sizes, a classifier frozen on no-cover traffic, and last- and penultimate-destination rules. Timing attacks add timing features, supervised and label-free gap removal, context-aware removal and joint decoding; Supplement S2.3 defines them. *Sel.* denotes validation selection; reported paired tests use these frozen choices. *Max* is the descriptive largest test count in a finite suite, not an upper bound on inference. *Exploratory* marks analyses designed after earlier test results were seen. Tests are exact paired McNemar tests with Holm correction within each declared family. Original intervals use 10,000 action-stratified fixture bootstraps conditional on fitted attacks and the designed workload; the added diagnostics report paired counts and tests without new bootstrap intervals.

Exploratory extensions. Feature ablations refit the original suite using the same splits and seeds. No-time removes timestamps, gaps, durations and elapsed-time features while retaining destinations, order, bytes and every candidate procedure. No-position replaces ordered byte slots with destination summaries and removes explicit position features and position-only rules, while retaining timing order. The eight timing contrasts and four position contrasts form separate correction families (Supplement S2.9).

The record-only extension uses the original encrypted records, five idle-gap eventizations and a protocol-aware eventizer. It retains every refitted baseline, including raw-record LR/RF, and adds context removal and joint decoding. TRAIN and validation call-role labels fit the detector and choose its threshold; validation task accuracy selects the final attack. TEST call truth enters only audits performed after predictions have been saved. Joint decoding removes at most two of the top ten candidate events; it is not a two-call budget. The eight selected-strong-minus-baseline comparisons form one exploratory family (Supplement S3.4).

5.3. An ideal candidate-label reference

Decoy-query obfuscation hides a query among candidates [67]. For planner cover, each candidate is a complete task. Consider the ideal observation consisting only of the unordered real and k shadow labels, all recovered perfectly. If the real chain is exchangeable with the shadows, its posterior label probability is $n_c/(k+1)$, where n_c counts chains labeled c. The modal-label guess succeeds with probability

$$P^*(k) = \frac{\mathbb{E}[\max_c n_c]}{k+1}.$$

This is optimal accuracy for that ideal label observation. With one shadow and six uniform classes, $P^*(1) = \frac{1}{6} + \frac{5}{6} \cdot \frac{1}{2} = \frac{7}{12}$. Table 7 gives the exact enumeration. Drawing distinct shadows from the other labels yields $1/(k+1)$ while preserving each shadow's uniform marginal.

TABLE 7. IDEAL CANDIDATE-LABEL REFERENCE $P^*(k)$ FOR k SHADOW PLANNERS AND SIX CLASSES, COMPUTED EXACTLY (EXPERIMENT/SHADOW_FLOOR.PY). IID: SHADOW LABELS DRAWN UNIFORMLY; DISTINCT: DRAWN WITHOUT REPLACEMENT FROM CLASSES OTHER THAN THE REAL ONE. TOKENS ARE NOMINAL PLANNER MULTIPLES. ONLY $k = 1$, IID, WAS RUN.

k	iid	distinct	Tokens×	Live result ($k = 1$)
0	1.000	–	1	
1	.583	.500	2	Max 70 / 64 of 120
2	.491	.333	3	
3	.456	.250	4	
4	.428	.200	5	
Test-split reference: 70.5/120 (21 matching labels)				

The six-class prior reference is $1/6$; $P^*(k)$ instead describes ambiguity when all candidate labels are known but roles are hidden. Actual traces may obscure labels or disclose roles, so $P^*(k)$ is neither a universal upper nor lower bound on trace inference. Reused fixtures and cache histories also limit exchangeability.

6. Experimental Setup

Workload and planners. Each fixture is a synthetic support ticket requesting one of six business actions over a state with three orders, a customer address, a compensation quota, an escalation status and a ledger. LangGraph [56] alternates model decisions with real tool calls to seven MCP Streamable HTTP services over TLS 1.3/HTTP/2, with request and response bodies padded to 4,096 and 16,384 bytes. Two fixed-weight planners, Qwen3:8B and Llama3.1:8B, run at temperature zero with weights pinned by layer hashes. Each planner executes 300 fixtures per condition: 120 training, 60 validation and 120 paired test fixtures, balanced over the six actions. The label is the requested action, and wrong or failed executions remain in every denominator.

Cover policies. All four cover arms execute cover on an isolated clone and hide its responses from the real planner; they implement different call-generation strategies (Figure 2). *Immediate* cover follows an honest response directly. *History* cover samples a training-only honest gap and service duration, waits in real time and holds the response for the sampled minimum; the *sequential* variant blocks the real chain and the *concurrent* variant overlaps real model inference. *Shadow* runs a complete second planner on an independent ticket whose action is drawn uniformly, and a fair draw decides which chain starts first. Calls to the same service serialize. The shadow also changes call count and composition. Controller positions are fixed before collection among slots after the ticket and up to just after the reference business action; 79/120 test fixtures omit that final slot. Actual issued-call coverage, which also depends on planning and concurrent dispatch, is measured separately in Section 7.1. All no-cover episodes are collected before the cover arms, whose order rotates across fixtures; prompt

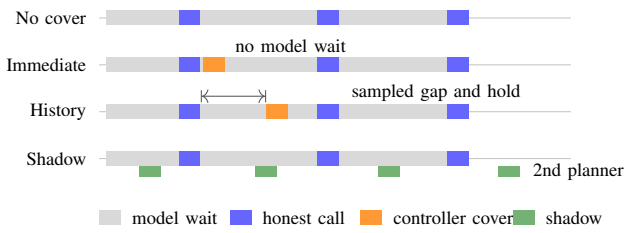


Figure 2. Dispatchers in the live study. Honest calls follow a model wait. Immediate cover follows the previous response without one, which label-free gap removal exploits. History-matched cover samples a training gap and response hold. Shadow calls come from a second planner, so their gaps are model waits too.

TABLE 8. STUDY MAP. EACH POPULATION ANSWERS ONE QUESTION AND KEEPS ITS OWN DENOMINATORS AND STATISTICAL ROLE.

Study	Population; observer	Question; role
Live timing (§7.1)	3,000 episodes + 600 shadow chains; call windows, TLS record times	Does the dispatcher expose cover? Primary families P, R, C
Feature/role diagnostics	Same episodes; call windows	What changes without time or explicit position? Exploratory refits and saved-prediction strata
Record-only (§7.2)	Same episodes; no TEST call boundaries	Original removal pair and later context/joint extension; eight exploratory contrasts
State control (§7.4; S1)	3,600 episodes; call windows, sizes only	Does goal completion miss collateral state? Primary state family

caches persist across arms. The five conditions give 3,000 main episodes and 600 shadow chains.

Studies. Table 8 lists the populations. The live timing study and its record-only reanalysis answer the privacy questions. A separate earlier state control on the same workload varies termination and feedback across six policies, including a deliberately harmful live-write arm, to test whether goal completion misses collateral state. All runs use one host over loopback with synthetic tools; Section 8 states what this does not test.

7. Results

7.1. A dispatch shortcut and residual trace context

Immediate cover exposes its short gap. A label-free two-component mixture over training gaps selects the short-gap component for removal, and a classifier trained on no-cover traffic decodes the rest. With supplied call windows it recognizes 112/120 and 108/120 tasks, against selected size-only attacks at 75 and 80 (Table 9). Both gains meet the replication rule after Holm correction ($p = 1.9 \times 10^{-8}$ and 4.3×10^{-6}). Independent TCP reassembly matches all 773,450 TLS record types, lengths and counts, and packet-capture clocks reproduce both scores. The mixture flags 181/182 and 182/182 held-out immediate decoys with no false flags (Supplement S2.7). This directly supports the short-gap mechanism in the measured planner loop.

Sampled waits leave recognizable call context. Sampling training gaps defeats the mixture: recognition falls to 53 and 14 (sequential) and 38 and 66 (concurrent). Context removal or joint decoding recognizes 93/120 and 104/120 sequential tasks against size-only attacks at 76 and 82, and 99/120 and 106/120 concurrent tasks against 66 and 68. Three comparisons survive Holm correction: Llama sequential ($p = .00078$) and both concurrent comparisons ($p < 10^{-4}$); Qwen sequential does not ($p = .0559$).

Removing time reduces residual recognition. The exploratory refitting ablation separates the contribution of time features from the original mixed-context comparison (Table 10). With destinations, order, bytes and every candidate procedure retained, removing time lowers concurrent recognition from 99 to 66/120 for Qwen and from 106 to 77/120 for Llama. Both survive Holm correction within the eight timing contrasts ($p = 2.37 \times 10^{-5}$ and 8.68×10^{-6}). Llama sequential falls from 104 to 80 ($p = 5.06 \times 10^{-5}$); Qwen sequential falls from 93 to 76 but does not pass correction ($p = .0931$). None of the four explicit-position ablations is significant in its separate family. The added tests measure the fitted suite's use of time, while the retained order information and changed position representation limit any conclusion about order itself. Supplement S2.9 gives every contrast and candidate score location.

An uncovered final action is not the whole explanation. We stratify saved predictions by whether cover starts after the last issued honest business call. Actual history-arm coverage is 37/120 and 47/120 for Qwen, and 13/120 and 40/120 for Llama, in sequential and concurrent dispatch. Within those covered strata, the selected adaptive attacks still recognize 29/37 and 39/47 Qwen tasks, and 11/13 and 37/40 Llama tasks. These observations rule out attributing all residual recognition to a final honest destination left alone, but the nonrandom strata do not isolate a placement effect (Supplement S2.10).

The achieved timing law also differs from its source draws. Sampled waits are minimums: Qwen's 40.5 ms mean hold becomes 100.4 ms sequential and 171.2 ms concurrent at the observer (Table 11). Llama's sequential mixture deletes 477/634 honest calls while detecting only 64/182 cover calls, explaining its 14/120 score; joint decoding on the same traces reaches 104/120. A failed removal model therefore leaves a separate residual-recognition question.

7.2. What survives without supplied call windows

Context decoding recovers task recognition from complete encrypted connections without TEST call windows. The input is unchanged: relative record times, directions, ciphertext lengths and destinations, including setup and teardown. Whole-episode connection association remains supplied. No test operation, role or HTTP stream assignment enters inference. The exploratory extension adds context removal and joint decoding on six fixed eventizations while

TABLE 9. JOINT EVALUATION OF THE FIVE LIVE POLICIES; 120 TEST FIXTURES PER ROW, SINGLE HOST OVER LOOPBACK, SYNTHETIC TOOLS. SEL. S/A: VALIDATION-SELECTED SIZE-ONLY / ADAPTIVE ATTACK (PRIMARY). MAX: LARGEST TEST COUNT IN THE FROZEN SUITE (DESCRIPTIVE). GMM: LABEL-FREE REMOVAL. REF.: UNIFORM SIX-CLASS REFERENCE 20/120, OR FOR SHADOW THE IDEAL CANDIDATE-LABEL REFERENCE ON THE REALIZED TEST SPLIT (TABLE 7), NOT A BOUND ON TRACE INFERENCE. BYTES: WHOLE-CONNECTION / CALL-WINDOW FACTORS. TOKENS INCLUDE BOTH PLANNERS. GOAL AND STATE ARE SEPARATE COUNTS FROM THIS COHORT; THE STATE-CONTROL COHORT (SUPPLEMENT S1) HAS ITS OWN COUNTS.

Planner	Policy	Sel. S/A	Max	GMM	Ref.	Bytes×	Tokens×	Goal/State
Qwen3:8B	No cover	117/117	117	–	20	1.00/1.00	1.00	113/114
Qwen3:8B	Immediate	75/112	112	112	20	1.15/1.33	1.00	111/112
Qwen3:8B	History, sequential	76/93	99	53	20	1.15/1.32	1.00	112/113
Qwen3:8B	History, concurrent	66/99	99	38	20	1.15/1.33	1.00	111/112
Qwen3:8B	Planner shadow	54/54	70	34	70.5	1.31/1.68	2.01	112/113
Llama3.1:8B	No cover	110/110	112	–	20	1.00/1.00	1.00	82/70
Llama3.1:8B	Immediate	80/108	108	108	20	1.13/1.27	1.01	83/67
Llama3.1:8B	History, sequential	82/104	104	14	20	1.13/1.26	1.01	84/67
Llama3.1:8B	History, concurrent	68/106	106	66	20	1.13/1.27	1.02	82/67
Llama3.1:8B	Planner shadow	64/64	64	43	70.5	1.35/1.71	1.99	82/67

TABLE 10. EXPLORATORY FEATURE ABLATIONS ON THE SAME TEST FIXTURES. FULL, NO-TIME AND NO-POSITION ARE VALIDATION-SELECTED TASK-RECOGNITION COUNTS /120. NO-TIME RETAINS DESTINATIONS, ORDER, CIPHERTEXT BYTES AND THE CANDIDATE SUITE. NO-POSITION CHANGES THE EXPLICIT-POSITION REPRESENTATION AND OMITS POSITION-ONLY RULES; TIMING ORDER REMAINS. p_T: FULL MINUS NO-TIME, HOLM WITHIN EIGHT TIMING CONTRASTS; p_P: FULL MINUS NO-POSITION, HOLM WITHIN FOUR CONTRASTS. ALL TESTS ARE TWO-SIDED.

Planner	Policy	Full	No-time	No-position	Time gain (pp)	p_T	p_P
Qwen	History, seq.	93	76	101	+14.2	0.0931	0.5059
Qwen	History, conc.	99	66	91	+27.5	2.4×10^{-5}	0.385
Llama	History, seq.	104	80	103	+20.0	5.1×10^{-5}	1
Llama	History, conc.	106	77	110	+24.2	8.7×10^{-6}	0.9614

TABLE 11. ACHIEVED HISTORY-BASED COVER TIMING, ALL SPLITS (455 COVER CALLS PER ROW). MEANS IN SECONDS. AN ANCHOR GAP IS MEASURED FROM THE ACTUAL PRIOR COMPLETED ANCHOR CALL TO COVER RECORD START. SAMPLED DURATIONS ARE MINIMUM SERVER HOLDS, WHILE ACHIEVED DURATIONS INCLUDE RECORD-LEVEL TRANSPORT AND SCHEDULING.

Model	Arm	Drawn gap	Achieved anchor gap	Drawn hold	Achieved duration
Qwen	History seq.	0.5198	0.5660	0.0405	0.1004
Qwen	History concurrent	0.5198	0.5772	0.0405	0.1712
Llama	History seq.	0.5865	0.6203	0.0768	0.1097
Llama	History concurrent	0.5865	0.6204	0.0768	0.1234

retaining all refitted baselines, including raw-record classifiers (Section 5.2).

Stronger decoders recover the controller signal. The selected strong family improves over its selected baseline in five of eight cells after Holm correction (Table 12). All four history comparisons improve significantly: 73 to 90 and 56 to 80/120 for Qwen, and 88 to 107 and 78 to 112/120 for Llama. The concurrent gains are 20.0 and 28.3 percentage points ($p = .00791$ and 8.61×10^{-9}). Immediate Qwen improves from 80 to 112 ($p = 2.85\times10^{-8}$). Llama's immediate increase from 90 to 102 is nonsignificant ($p = .1503$). Both shadow contrasts are negative and nonsignificant; validation selection across all candidates retains the 69/120 baseline in each.

Pipeline failure and privacy are different results. The earlier short-gap mixture scores 69/120 and 86/120 on immediate cover, against baselines of 80 and 90 (original Holm $p = .025$ and $.54$). Those results and the later protocol-eventizer diagnostic are preserved in Supplement S3.5. The added decoders instead use learned event context and joint task evidence. Their selected immediate eventizations reach call-boundary IoU-.5 F1 of .571 and .632, and event-role F1 of .992 and 1.000; Supplement S3.4 separates these endpoints and records mixed events. A weak segmentation/removal pipeline therefore cannot establish protection when adapted decoding still recovers tasks.

The strongest tested counts span 76 candidates and remain descriptive. Training and validation role annotations support detector fitting; test call truth is used only in post-prediction audits. Removing two inferred events differs from removing two supplied calls, and the record-only input also includes setup and teardown. These representation and suite differences prevent a pure causal comparison of boundary access. The demonstrated result is recovery from associated encrypted records, with task discovery in an unassociated stream still outside this experiment.

TABLE 12. EXPLORATORY ATTACKS WITHOUT TEST CALL WINDOWS; EPISODE ASSOCIATION REMAINS SUPPLIED. COUNTS /120. B: VALIDATION-SELECTED BASELINE, INCLUDING ALL RAW-RECORD CANDIDATES. J: VALIDATION-SELECTED CONTEXT-REMOVAL/JOINT-DECODING CANDIDATE. ALL: VALIDATION SELECTION ACROSS ALL 76 CANDIDATES. MAX: DESCRIPTIVE TEST MAXIMUM. DIFFERENCE AND HOLM p COMPARE J WITH B IN ONE EIGHT-TEST FAMILY. BOTH SHADOW CELLS SELECT THE BASELINE IN ALL.

Planner	Policy	B	J	All	Max	J−B (pp)	Holm p
Qwen	Immediate	80	112	112	114	+26.7	2.9×10^{-8}
Qwen	History, seq.	73	90	90	99	+14.2	0.0182
Qwen	History, conc.	56	80	80	95	+20.0	0.0079
Qwen	Shadow	69	60	69	73	-7.5	0.5057
Llama	Immediate	90	102	102	112	+10.0	0.1503
Llama	History, seq.	88	107	107	111	+15.8	0.0079
Llama	History, conc.	78	112	112	112	+28.3	8.6×10^{-9}
Llama	Shadow	69	64	69	69	-4.2	0.5057

7.3. Recognition and work under planner shadows

The strongest tested call-window attack, a size/destination RF, recognizes 70/120 Qwen and 64/120 Llama shadow tasks; validation-selected attacks score 54 and 64. The expanded 76-candidate record-only suite retains maxima of 73 and 69, with overall validation selections at 69 and 69. These are residuals of the complete shadow policy.

The shadow uses 8.14 and 10.46 total tool calls per fixture against controller cover's 5.6 and 6.8, changes their composition to a complete second task, and uses 2.01 and 1.99 times the model tokens. Real-chain time rises from 2.91/3.33 s to 4.80/3.94 s; call-window bytes rise 1.68/1.71 times. This quantifies the complete policy's trade-off. Its second business task also changes terminal coverage: later shadow business calls occur in 74/120 Qwen and 63/120 Llama fixtures. Isolating online planner generation requires a controller replay with matched calls, composition and terminal coverage.

The ideal observer in Section 5.3 knows both candidate labels but no real-chain identity. With 21 matching pairs among 120 fixtures, its reference is 70.5/120; this neither bounds actual trace inference nor establishes role indistinguishability. Exact saved-prediction analysis gives RF task recognition of 49/99 and 44/99 on different-label pairs. The chain attack recovers the entire real chain in 29/120 and 32/120 fixtures. First-role prediction, exact-chain recovery and task recognition remain distinct endpoints; Supplement S2.11 gives the role tests and start-order baseline. The original shadow equivalence family compares identical selected attacks and supplies no role-indistinguishability evidence.

7.4. Business outcomes from the same runs

The live study records goal, state and cost on the same executions as traffic. All cover runs on isolated clones. Goal/state counts stay within three fixtures of no cover (Table 9); honest paths match no cover on 115/120 Qwen and 103/120 Llama shadow fixtures. Llama's baseline goal/state counts, 82/70 of 120, define a limited executor baseline rather than a production reliability target.

A separate state-control cohort tests whether these endpoints distinguish collateral effects. Proxy, read and hidden-isolated cover keep Qwen state correctness at 115–116/120. Deliberate live writes reduce it to 7/120 while 102/120 goals complete, including 95 with collateral errors. Llama has 3/120 intact and 73 of 76 completed tasks with collateral errors. Both integrity reductions survive Holm correction ($p = 3.7 \times 10^{-32}$ and 3.0×10^{-19}; Supplement S1). These constructed cases demonstrate endpoint sensitivity; their separate counts estimate no production damage rate.

7.5. A worked screening example

Table 13 uses illustrative tolerances chosen after measurement. It is a descriptive worksheet, not a validated deployment decision. Controller maxima exceed its 60% point threshold in both observation representations; shadow's displayed call-window count intervals straddle it. A decision study must fix selection and uncertainty procedures in advance and cover every mandatory outcome, including paired goal/state changes and costs.

The sample has limited resolution near 60%. An attack performing exactly at the iid reference $7/12 \approx .583$ needs about 3,400 fixtures before its 95% Clopper–Pearson upper limit falls below 60% (1,983/3,400 gives .5999); at the distinct-label reference of .5, 60/120 already gives .593. Neither calculation is a powered role-equivalence study. Distinct-label sampling changes the ideal reference and remains unmeasured.

The worksheet now includes the expanded record-only suite, whose history maxima are 95–112/120. Different representations and attack families still prevent reading its difference from call-window results as the value of boundaries. Observed mean costs are below the worksheet budgets; complete joint uncertainty is unassessed. Figure 3 shows the measured call-window policy points; Table 7 gives the ideal label references separately.

8. Discussion

What the measurements add. Where an implementer generates a tool call becomes part of its traffic signature. Immediate cover bypasses the model wait and exposes a short-gap feature. Marginal sampled waits remove that shortcut while leaving temporal context useful: the exploratory

TABLE 13. DESCRIPTIVE SCREENING WORKSHEET. TOLERANCES WERE CHOSEN AFTER MEASUREMENT: RECOGNITION $\leq 60\%$, GOAL/STATE POINT DIFFERENCES AT MOST 3/120 BELOW NO COVER, MODEL TOKENS $\leq 2.5\times$, CALL BYTES $\leq 2\times$ AND MEAN REAL-CHAIN LATENCY $\leq 2\times$. CW MAX IS THE LARGEST TESTED COUNT WITH A NOMINAL 95% CLOPPER–PEARSON COUNT INTERVAL, NOT AN INTERVAL FOR ALL-ATTACKER PERFORMANCE OR A PROSPECTIVELY SELECTED ATTACK. GOAL/STATE AND COSTS ARE DESCRIPTIVE. ABOVE AND UNRESOLVED REFER ONLY TO THE WORKSHEET; NO POLICY HAS A VALIDATED JOINT PASS. RO MAX USES THE 76-CANDIDATE EXPLORATORY RECORD-ONLY SUITE; THE OBSERVATION REPRESENTATIONS AND ATTACK SUITES DIFFER.

Planner	Policy	CW Max [95% CI]	RO Max	ΔG/ΔS	Tok.$\times$	Bytes$\times$	Lat.$\times$	CW screening	RO screening
Qwen3:8B	Immediate	112 [.87, .97]	114	$-2/-2$	1.00	1.33	1.23	Above: inference	Above: inference
Qwen3:8B	History, sequential	99 [.75, .89]	99	$-1/-1$	1.00	1.32	1.47	Above: inference	Above: inference
Qwen3:8B	History, concurrent	99 [.75, .89]	95	$-2/-2$	1.00	1.33	1.14	Above: inference	Above: inference
Qwen3:8B	Planner shadow	70 [.49, .67]	73	$-1/-1$	2.01	1.68	1.65	Unresolved: inference	Unresolved: inference
Llama3.1:8B	Immediate	108 [.83, .95]	112	$+1/-3$	1.01	1.27	0.88	Above: inference	Above: inference
Llama3.1:8B	History, sequential	104 [.79, .92]	111	$+2/-3$	1.01	1.26	1.15	Above: inference	Above: inference
Llama3.1:8B	History, concurrent	106 [.81, .94]	112	$0/-3$	1.02	1.27	0.84	Above: inference	Above: inference
Llama3.1:8B	Planner shadow	64 [.44, .63]	69	$0/-3$	1.99	1.71	1.18	Unresolved; state at point limit	Unresolved: inference

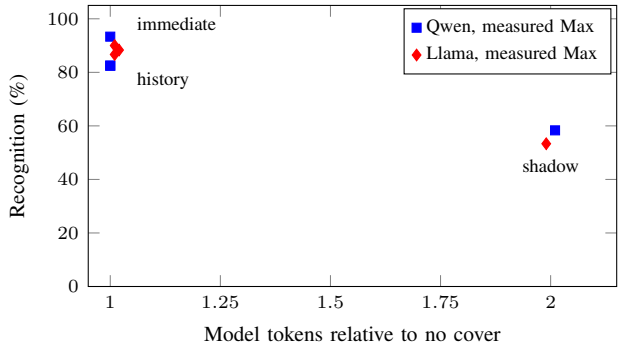


Figure 3. Measured recognition and model work for the four cover policies. Points are strongest tested call-window attacks; overlapping controller points retain their values. Shadow changes call count and composition as well as model work. Ideal candidate-label values are shown separately in Table 7.

refitting ablation loses recognition without time in three of four history cells. Context decoding also raises record-only recognition in five of eight comparisons. Together these results direct evaluation toward the dependencies between calls, beyond the marginal gap used by the first remover.

The shadow policy changes that sequence at a measured cost. Lower recognition and approximately doubled tokens describe a complete policy point; choosing online generation over controller replay requires a matched comparison. Execution authority is a second design choice. A cooperating provider can synthesize a response, use a sandbox, or execute a business operation. Those choices have different state and resource obligations even when they seek similar traces.

Measurements that distinguish the remaining explanations. The ablations measure feature use, and the record-only attacks measure recovery in a different representation. Controlled replays can now hold calls, composition and terminal coverage fixed while changing online dispatch. Interleaved no-cover controls can separate policy costs from collection phase. A service-capability comparison can remove response padding and holds while keeping safe synthetic states, and a joint model/tool observer can test whether each tool call follows a model exchange. Each question needs its own matched intervention.

Scope and uncertainty. This is a single-host study with synthetic tools, two fixed 8B models and cooperating services (Table 5). Same-service calls serialize. Fixtures are paired across conditions and reused across studies; request phrasings recur across splits. Cross-host deployment, unseen templates, unrestricted multiplexing, recipient arguments, non-cooperating services and simultaneous model-link observation are unmeasured; the last two are the common deployment cases. Loopback removes network noise; longer remote inference could strengthen the dispatch contrast. Neither extrapolation is estimated here.

Intervals condition on fitted attacks and reused workloads. The added diagnostics were designed after earlier results were known, with code and families fixed before their new scores; they are exploratory, not a new cohort or confirmation study. Test maxima describe the suite, not a population optimum. The worksheet uses post-measurement thresholds and lacks full joint uncertainty. No-cover episodes were collected first and caches persist across arms, so cover-versus-none comparisons of cost, goal and state include phase effects; Llama's immediate and concurrent latencies of $0.88\times$ and $0.84\times$ (Table 13) are not causal speedup estimates. Within-arm attack and representation comparisons do not compare those collection phases. The separate state stress cohort illustrates endpoint sensitivity rather than harm prevalence. Appendix A's four privacy settings come from two sources and motivate the joint record without estimating a field-wide rate.

9. Related Work

Agent side channels. Section 2 and Appendix A cover Agent communication attacks. WRA tests research decoys and destination visibility [2]; the communication-graph study tests transformations and live OHTTP [3]. Serving-side studies examine streams, cache sharing and token counts [20], [12], [11], [13], [21], [14]. Our controlled

study follows calls into TLS traces while measuring work under each policy.

Decoy-query obfuscation. TrackMeNot generates periodic queries [63]; GooPIR combines real and decoy candidates [67]. Classifiers and systematic studies show how generated queries remain distinguishable [64], [65], [68]. This generator principle is the starting point here. We examine its interaction with model-decision latency and tool authority using actual dispatch, two observation representations and matched business/resource outcomes. The candidate-label reference is the deniability calculation for complete task labels.

Traffic defenses and Agent evaluation. WF supplies adapted-attack, observer and overhead requirements [15], [16], [32], [17], [18], [66], [69], [62]; smart-home shaping terminates cover at a tunnel [34]. Agent information-flow controls govern tool authority [57], [58], [59]; SPILLage examines behavioral oversharing [27]. Environment outcomes in τ-bench and AgentDojo [60], [61] motivate our state endpoint. We bind those endpoints to protected traffic, complementing lifecycle and attack-surface surveys [53], [54], [55].

10. Ethics and Availability

Ethics. Inputs are synthetic, captures are restricted to the experiment, and state-stress writes affect isolated fixtures. The hosted pilot preserved in the evidence archive used account-holder authorization; provider research terms remain unverified, and it supports none of the present main results. Traffic inference can enable profiling; our measurements target synthetic tickets. AI tools assisted extraction, implementation and writing; observations and analyses are preserved. Credentials, hosted transcripts and captures, and third-party full texts are excluded from distribution.

Availability. The accompanying review artifact provides manuscript sources, versioned collection and analysis implementations, fixture generation, frozen plans and historical timing pools, aggregate tests and costs, the ideal-reference computation, and the source-located background matrix. The added feature-ablation and record-context release includes every candidate's aggregate validation/test count, selected contrasts, event audits and an aggregate arithmetic checker. Its index distinguishes those checks from full-data replication. Per-episode records, individual timestamps and predictions, captures and TLS keys remain local. Independent refitting, feature reconstruction and per-fixture role checks require those withheld inputs. No public release of them is promised.

11. Conclusion

Cover generation and execution must be evaluated together. In the measured Agent loop, controller insertion bypasses a model wait and exposes a short-gap signature. Sampled waits remove that shortcut, yet feature ablation shows useful time dependence in three of four history cells. Context removal and joint decoding recover recognition from associated TLS records without supplied test call windows, improving over baselines in five of eight cells. Shadows remain a distinct complete-policy trade-off with more calls and model work. Recording inference, goals, business state and cost on matched executions connects these trace dependencies to the authority and resources a cover policy consumes.

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Appendix A. Background Evidence Scope

The companion matrix contains forty-four settings from fifteen Agent communication studies. Inclusion requires an LLM to select actions with external feedback and a communication-side inference or evaluated protection. Client, model, tool and coordination interfaces are eligible; transit, recipient and active access remain separate. The bounded window covers first public submissions from 1 January 2024 through 9 November 2026. The evidence archive preserves the exact retrieval, screening, source-version and coding record (its background-extraction report, Sections A and S4–S6).

The settings comprise twenty-two transit observations, ten recipient settings, eleven active settings and one inferred-preference setting. Their source-defined scores are background evidence, not a common benchmark. The four query/workflow privacy settings cited in Section 2.4 come from two sources. The source-located matrix and archived coding record permit inspection of that narrow comparison without treating it as a field-wide frequency.